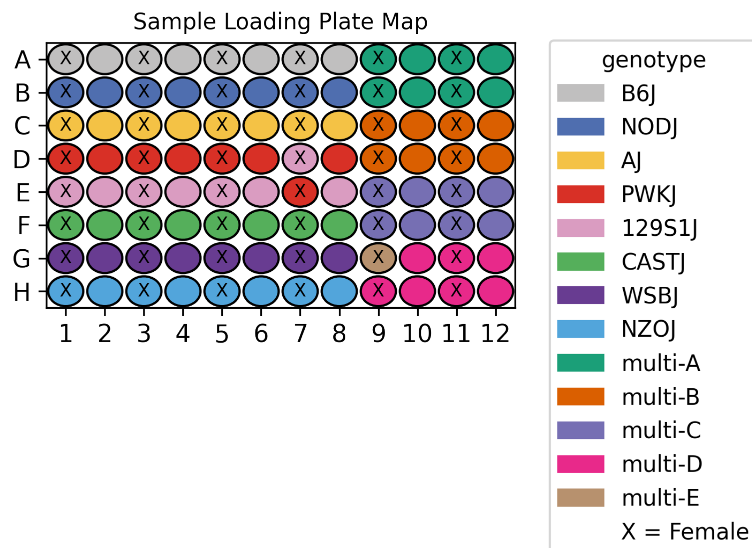
UCI/Caltech IGVF Dataset Pipeline Report

	igvf_005
Kit	WT_mega
Chemistry	Parse v2
Sample type	nuclei
Genotypes	B6J NODJ AJ PWKJ 129S1J CASTJ WSBJ NZOJ
Tissues	Liver DiencephalonPituitary
Subpool	Subpool_3
Measurment Set	IGVFDS4723VQYS
Exome capture subpool	No
Expected nuclei	67,000



1. Sample multiplexing

Liver was the main tissue on the plate while DiencephalonPituitary was multiplexed.

Table 1: Genotype multiplexing key

Pair	Multiplexed Genotype 1	Multiplexed Genotype 2
multi-A	B6J	NODJ
multi-B	AJ	PWKJ
multi-C	129S1J	CASTJ
multi-D	NZOJ	WSBJ
multi-E	CASTJ	WSBJ

2. Alignment

A total of 1,486,520,139 reads were aligned using GENCODE vM32 with mm39 assembly.

Table 2: Pseudoalignment stats

Subpool	Total reads	Total UMIs	Fraction duplicated UMIs	Total pseudoaligned	Pct. pseudoaligned
Subpool_3	1,486,520,139	747,680,637	0.291	1,054,683,153	70.9

3. Cell recovery

A total of 242,660 nuclei in Subpool_3 have >=200 UMIs and 68,099 nuclei have >= 500 UMIs.

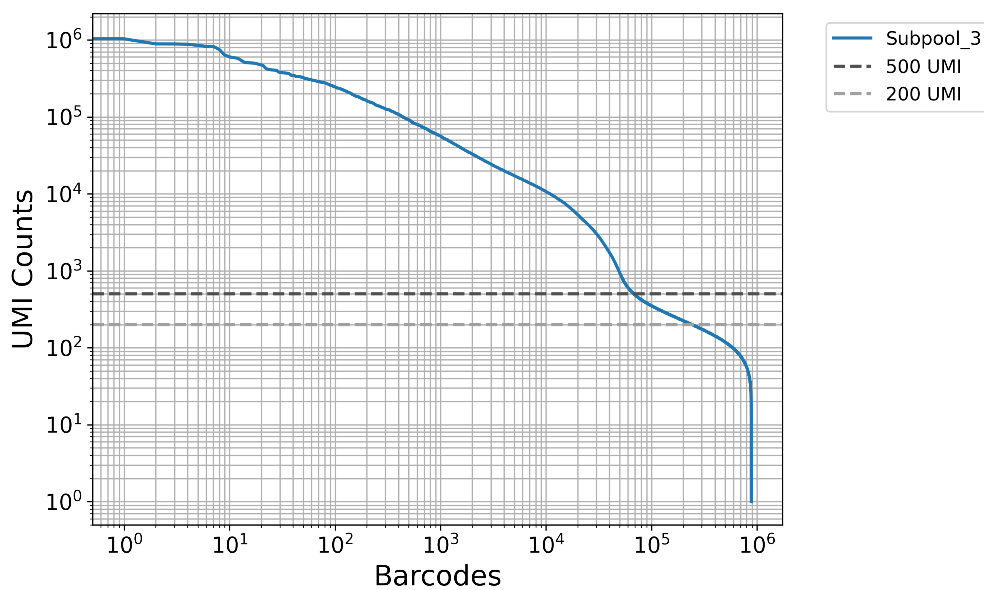


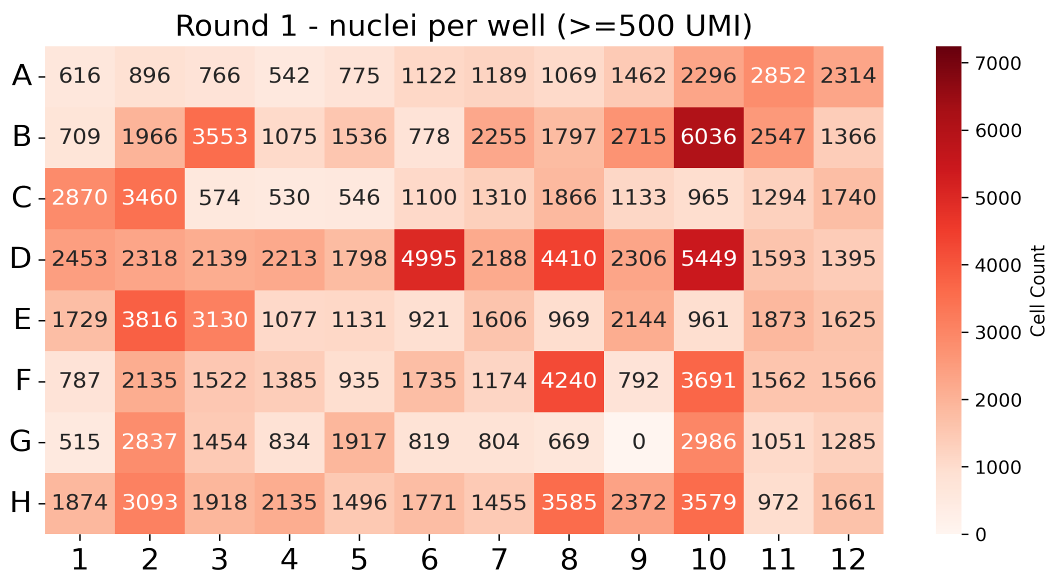
Table 3: QC stats for nuclei >=200 UMI

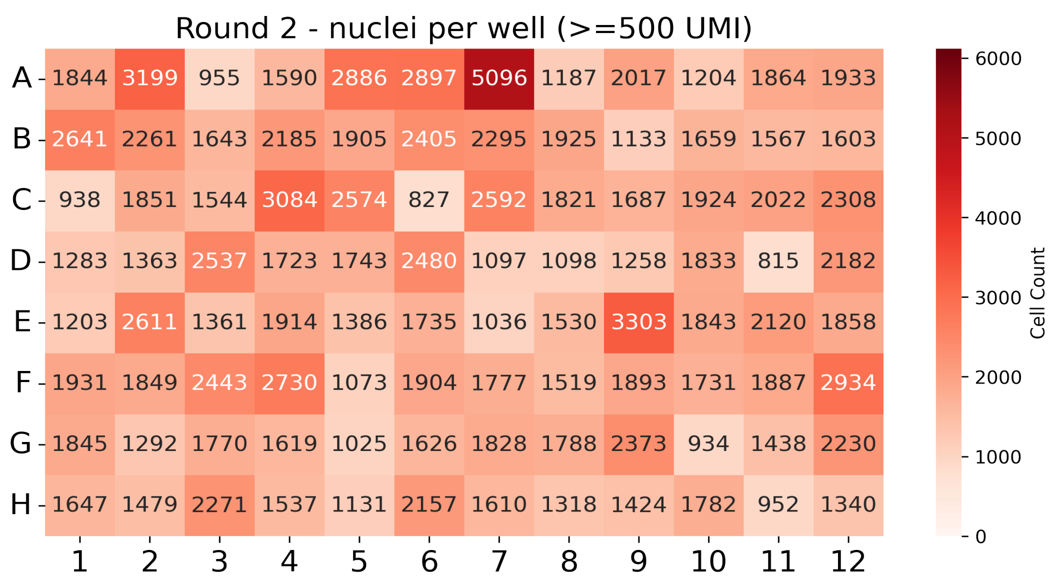
Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_3	242,660	307	2,154.7	267	751.9

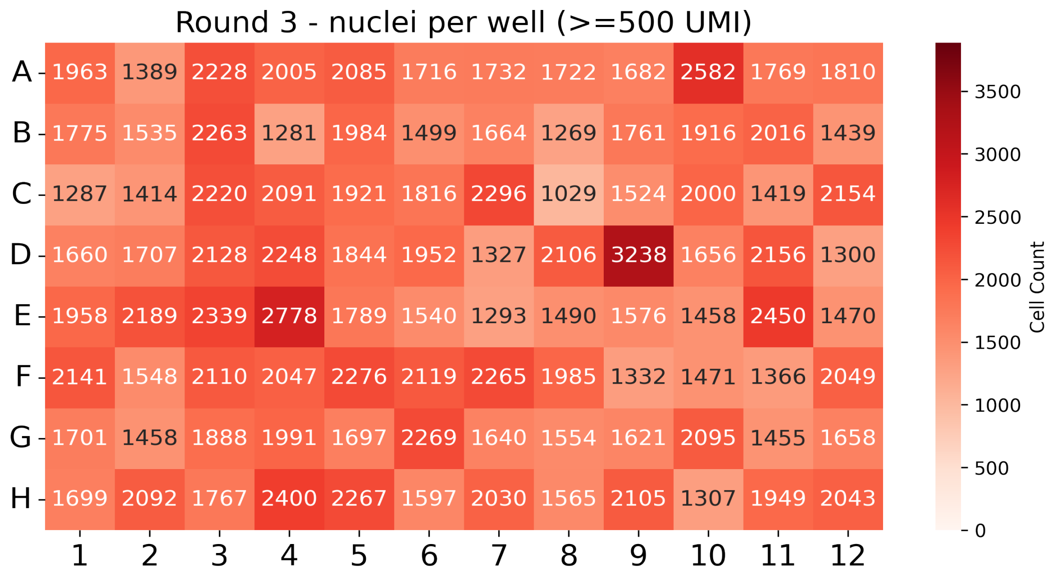
Table 4: QC stats for nuclei >=500 UMI

Subpool	Number of nuclei	Median UMIs	Mean UMIs	Median genes	Mean genes
Subpool_3	68,099	2,424	6,951.0	1,431	2,044.4

In this combinatorial barcoding assay, nuclei are barcoded in 96-well plates over three rounds. The first round assigns sample-specific barcodes, while the next two are applied to the pooled mix. Even distribution in the first round is important to ensure consistent sample representation.







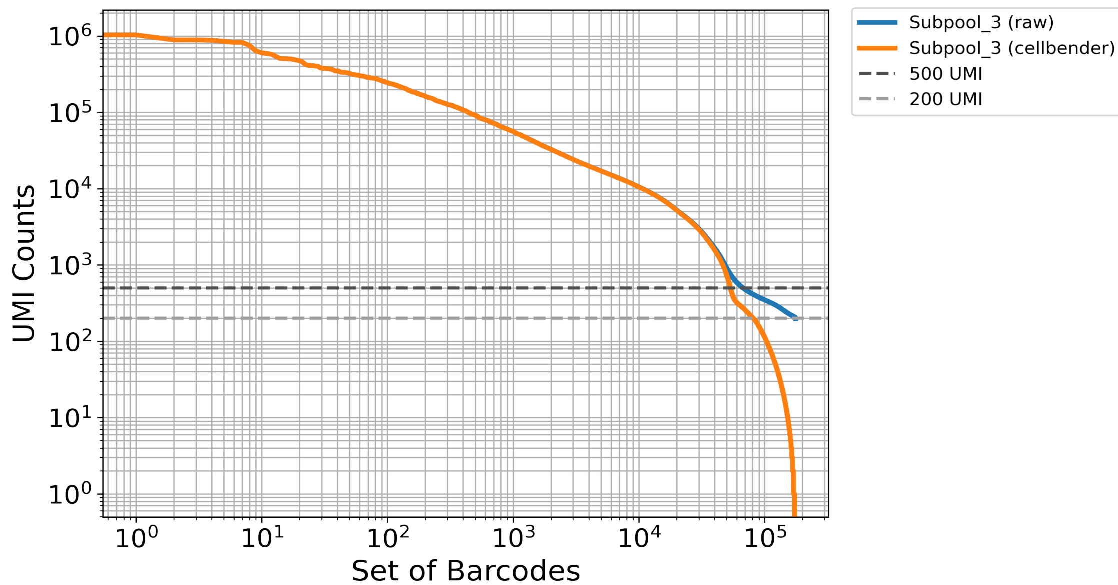
4. Background removal

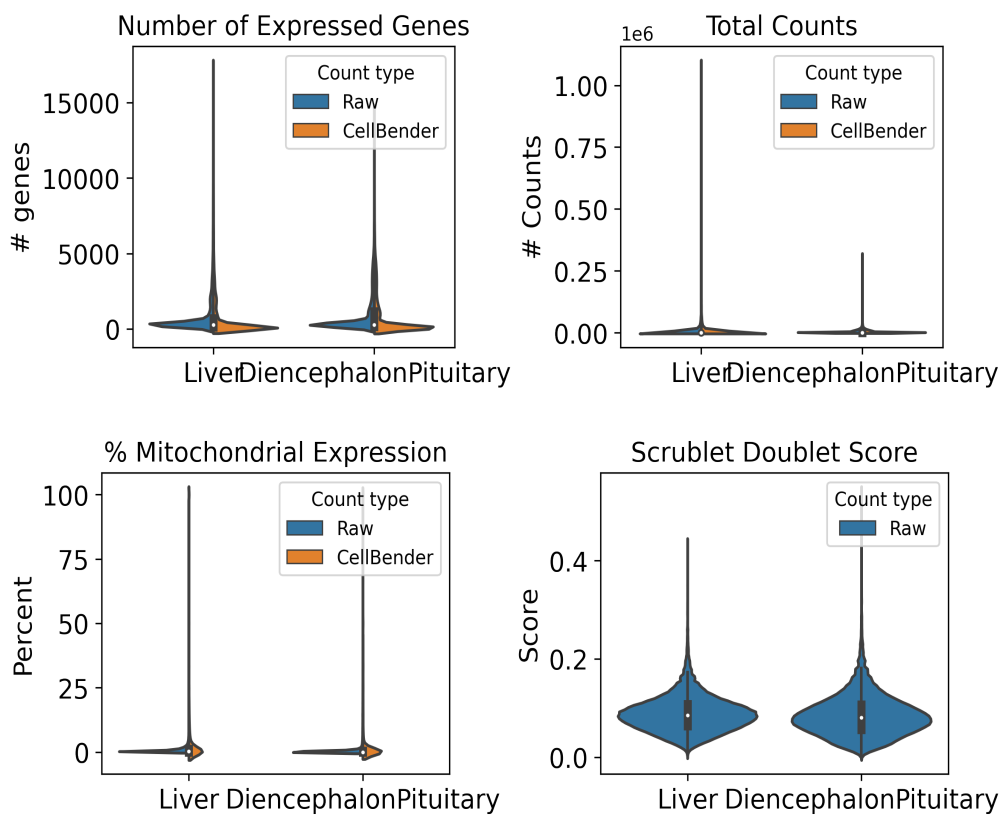
Table 5: Cellbender parameters

Subpool	droplets_included	learning_rate	expected_cells
Subpool_3	250000	5e-05	67000

Table 6: Cellbender stats

Subpool	Percent counts removed	Found nuclei	Mean denoised counts
Subpool_3	6.7	178,319	2,655.6





5. Barcode sample map

The first round corresponds to sample barcoding. In Parse Bio assays, two barcodes are included in the first round: a random hexamer (R) and oligo-dT (T) barcode per well (parse barcode type). This reduces 3-prime bias and allows for the capture of full-length transcripts. These sample level barcodes can be found in the barcode-1 region in the seqspec, and begin at position 16 (0-based) in the full 24-nt cell barcode. For the WT_mega kit, samples are distributed across 96 wells in round 1. The following table includes our lab sample ID as sample descriptions and IGVF sample accessions.

Table 7: Barcode sample map

barcode	sample accession	sample description	position	parse barcode type	well
CATCATCC	IGVFSM2609IWQV	016_B6J_10F_05	16	R	A1
CATTCCTA	IGVFSM2609IWQV	016_B6J_10F_05	16	T	A1
CTGCTTTG	IGVFSM5795GRDC	017_B6J_10M_05	16	R	A2
CTTCATCA	IGVFSM5795GRDC	017_B6J_10M_05	16	T	A2
CTAAGGGA	IGVFSM9969QCIJ	018_B6J_10F_05	16	R	A3
CCTATATC	IGVFSM9969QCIJ	018_B6J_10F_05	16	T	A3
GCTTATAG	IGVFSM6046EOEX	019_B6J_10M_05	16	R	A4
ACATTTAC	IGVFSM6046EOEX	019_B6J_10M_05	16	T	A4
TCTGATCC	IGVFSM1078KTYU,IGVFSM8134HPUH	092_CASTJ_10F_01	16	R	A5
ACTTAGCT	IGVFSM1078KTYU,IGVFSM8134HPUH	092_CASTJ_10F_01	16	T	A5
TCTCTTGG	IGVFSM9376MAOE	021_B6J_10M_05	16	R	A6
CCAATTCT	IGVFSM9376MAOE	021_B6J_10M_05	16	T	A6
CAATTTCC	IGVFSM4625KNRU	024_B6J_10F_05	16	R	A7
GCCTATCT	IGVFSM4625KNRU	024_B6J_10F_05	16	T	A7
AGTCTCTT	IGVFSM9141WALR	025_B6J_10M_05	16	R	A8
ATGCTGCT	IGVFSM9141WALR	025_B6J_10M_05	16	T	A8
TGCTGCTC	IGVFSM4465YOJN,IGVFSM5984CJUU	016_B6J_10F_01	16	R	A9
CATTTACA	IGVFSM4465YOJN,IGVFSM5984CJUU	016_B6J_10F_01	16	T	A9
TGCTGCTC	IGVFSM3183PSGJ,IGVFSM6418OJGM	066_NODJ_10F_01	16	R	A9
CATTTACA	IGVFSM3183PSGJ,IGVFSM6418OJGM	066_NODJ_10F_01	16	T	A9
GTATTTCC	IGVFSM5230YSVJ,IGVFSM6071JHCX	017_B6J_10M_01	16	R	A10
ACTCGTAA	IGVFSM5230YSVJ,IGVFSM6071JHCX	017_B6J_10M_01	16	T	A10
GTATTTCC	IGVFSM1910QUCB,IGVFSM6194RATY	067_NODJ_10M_01	16	R	A10
ACTCGTAA	IGVFSM1910QUCB,IGVFSM6194RATY	067_NODJ_10M_01	16	T	A10
TTCCTGTG	IGVFSM5188HZIG,IGVFSM9214NKTQ	018_B6J_10F_01	16	R	A11
CCTTTGCA	IGVFSM5188HZIG,IGVFSM9214NKTQ	018_B6J_10F_01	16	T	A11
TTCCTGTG	IGVFSM4182XRQE,IGVFSM8455TNVO	068_NODJ_10F_01	16	R	A11

CCTTTGCA	IGVFSM4182XRQE,IGVFSM8455TNVO	068_NODJ_10F_01	16	T	A11
GCTGCTTC	IGVFSM0166IBCH,IGVFSM6699LJHC	019_B6J_10M_01	16	R	A12
ACTCCTGC	IGVFSM0166IBCH,IGVFSM6699LJHC	019_B6J_10M_01	16	T	A12
GCTGCTTC	IGVFSM6842OMYX,IGVFSM8873UKSN	069_NODJ_10M_01	16	R	A12
ACTCCTGC	IGVFSM6842OMYX,IGVFSM8873UKSN	069_NODJ_10M_01	16	T	A12
TATGTGTC	IGVFSM7322GJHR	066_NODJ_10F_05	16	R	B1
ATTTGGCA	IGVFSM7322GJHR	066_NODJ_10F_05	16	T	B1
CAATTCTC	IGVFSM1637CAZY	067_NODJ_10M_05	16	R	B2
TTATTCTG	IGVFSM1637CAZY	067_NODJ_10M_05	16	T	B2
TGGTCTCC	IGVFSM8221YADP	068_NODJ_10F_05	16	R	B3
TCATGCTC	IGVFSM8221YADP	068_NODJ_10F_05	16	T	B3
GCTCTTTA	IGVFSM2794WEMQ	069_NODJ_10M_05	16	R	B4
CATACTTC	IGVFSM2794WEMQ	069_NODJ_10M_05	16	T	B4
GCTGCATG	IGVFSM1557PJAI	070_NODJ_10F_05	16	R	B5
CCGTTCTA	IGVFSM1557PJAI	070_NODJ_10F_05	16	T	B5
ACTCATTT	IGVFSM5227MWGM	071_NODJ_10M_05	16	R	B6
GCTTCATA	IGVFSM5227MWGM	071_NODJ_10M_05	16	T	B6
AGTCTTGG	IGVFSM8370FQBH	072_NODJ_10F_05	16	R	B7
CTCTGTGC	IGVFSM8370FQBH	072_NODJ_10F_05	16	T	B7
GGTTCTTC	IGVFSM3979KZZX	073_NODJ_10M_05	16	R	B8
CCCTTATA	IGVFSM3979KZZX	073_NODJ_10M_05	16	T	B8
TCATGTTG	IGVFSM3470VQGS,IGVFSM8528YUHU	020_B6J_10F_01	16	R	B9
ACTGCTCT	IGVFSM3470VQGS,IGVFSM8528YUHU	020_B6J_10F_01	16	T	B9
TCATGTTG	IGVFSM0017ZTCP,IGVFSM1602MMIE	070_NODJ_10F_01	16	R	B9
ACTGCTCT	IGVFSM0017ZTCP,IGVFSM1602MMIE	070_NODJ_10F_01	16	T	B9
ATTTTGCC	IGVFSM7323JYDH,IGVFSM8952OBMF	021_B6J_10M_01	16	R	B10
CTCTAATC	IGVFSM7323JYDH,IGVFSM8952OBMF	021_B6J_10M_01	16	T	B10
ATTTTGCC	IGVFSM6636PLRM,IGVFSM8886HJDO	071_NODJ_10M_01	16	R	B10
CTCTAATC	IGVFSM6636PLRM,IGVFSM8886HJDO	071_NODJ_10M_01	16	T	B10
CTTCTGTA	IGVFSM1381VQIA,IGVFSM8656OHKM	024_B6J_10F_01	16	R	B11
ACCCTTGC	IGVFSM1381VQIA,IGVFSM8656OHKM	024_B6J_10F_01	16	T	B11
CTTCTGTA	IGVFSM0442MPEV,IGVFSM9173JXFM	072_NODJ_10F_01	16	R	B11
ACCCTTGC	IGVFSM0442MPEV,IGVFSM9173JXFM	072_NODJ_10F_01	16	T	B11
GTCCATCT	IGVFSM1156KMPC,IGVFSM4934KHPJ	025_B6J_10M_01	16	R	B12
ATCTTAGG	IGVFSM1156KMPC,IGVFSM4934KHPJ	025_B6J_10M_01	16	T	B12

GTCCATCT	IGVFSM0489RZXG,IGVFSM8174YDBO	073_NODJ_10M_01	16	R	B12
ATCTTAGG	IGVFSM0489RZXG,IGVFSM8174YDBO	073_NODJ_10M_01	16	T	B12
GCTATCTC	IGVFSM7760MHIK	026_AJ_10F_05	16	R	C1
CATGTCTC	IGVFSM7760MHIK	026_AJ_10F_05	16	T	C1
TAGTTTCC	IGVFSM8587LUPL	029_AJ_10M_05	16	R	C2
TCATTGCA	IGVFSM8587LUPL	029_AJ_10M_05	16	T	C2
TCCATTAT	IGVFSM8501AGYE	028_AJ_10F_05	16	R	C3
ACACCTTT	IGVFSM8501AGYE	028_AJ_10F_05	16	T	C3
AGGATTAA	IGVFSM9889OUOI	031_AJ_10M_05	16	R	C4
AATTTCTC	IGVFSM9889OUOI	031_AJ_10M_05	16	T	C4
AATCTTTC	IGVFSM9122CTMU	030_AJ_10F_05	16	R	C5
ATTCATGG	IGVFSM9122CTMU	030_AJ_10F_05	16	T	C5
GTCATATG	IGVFSM8488GXLP	033_AJ_10M_05	16	R	C6
ACTTTACC	IGVFSM8488GXLP	033_AJ_10M_05	16	T	C6
GTGCTTCC	IGVFSM3749AQYK	032_AJ_10F_05	16	R	C7
CTTCTAAC	IGVFSM3749AQYK	032_AJ_10F_05	16	T	C7
ATGTGTTG	IGVFSM2286VGQO	035_AJ_10M_05	16	R	C8
CTATTTCA	IGVFSM2286VGQO	035_AJ_10M_05	16	T	C8
CCATCTTG	IGVFSM2617YKKQ,IGVFSM8677EQEG	026_AJ_10F_01	16	R	C9
TCTCATGC	IGVFSM2617YKKQ,IGVFSM8677EQEG	026_AJ_10F_01	16	T	C9
CCATCTTG	IGVFSM5417LNGC,IGVFSM6924RRKD	076_PWKJ_10F_01	16	R	C9
TCTCATGC	IGVFSM5417LNGC,IGVFSM6924RRKD	076_PWKJ_10F_01	16	T	C9
TACTGTCT	IGVFSM1458SAIZ,IGVFSM8548SQFG	029_AJ_10M_01	16	R	C10
ATCCTTAC	IGVFSM1458SAIZ,IGVFSM8548SQFG	029_AJ_10M_01	16	T	C10
TACTGTCT	IGVFSM4103LTZB,IGVFSM5500OUCV	077_PWKJ_10M_01	16	R	C10
ATCCTTAC	IGVFSM4103LTZB,IGVFSM5500OUCV	077_PWKJ_10M_01	16	T	C10
TTCATCGC	IGVFSM4796DKJO,IGVFSM4881USYE	028_AJ_10F_01	16	R	C11
TAAATATC	IGVFSM4796DKJO,IGVFSM4881USYE	028_AJ_10F_01	16	T	C11
TTCATCGC	IGVFSM2146SQAW,IGVFSM9111MIVH	078_PWKJ_10F_01	16	R	C11
TAAATATC	IGVFSM2146SQAW,IGVFSM9111MIVH	078_PWKJ_10F_01	16	T	C11
ACTGTGGG	IGVFSM3351WODP,IGVFSM6893IKIH	031_AJ_10M_01	16	R	C12
TTACCTGC	IGVFSM3351WODP,IGVFSM6893IKIH	031_AJ_10M_01	16	T	C12
ACTGTGGG	IGVFSM3536WDNR,IGVFSM3744QPBB	079_PWKJ_10M_01	16	R	C12
TTACCTGC	IGVFSM3536WDNR,IGVFSM3744QPBB	079_PWKJ_10M_01	16	T	C12
TCTGTGCC	IGVFSM8222BTFH	076_PWKJ_10F_05	16	R	D1

CACTTTCA	IGVFSM8222BTFH	076_PWKJ_10F_05	16	T	D1
TCAATCTC	IGVFSM6195ZDYB	077_PWKJ_10M_05	16	R	D2
CACCTTTA	IGVFSM6195ZDYB	077_PWKJ_10M_05	16	T	D2
GTCCTCTG	IGVFSM0598HJAL	078_PWKJ_10F_05	16	R	D3
CTGACTTC	IGVFSM0598HJAL	078_PWKJ_10F_05	16	T	D3
TTACATTC	IGVFSM7868PMHQ	079_PWKJ_10M_05	16	R	D4
CATTTGGA	IGVFSM7868PMHQ	079_PWKJ_10M_05	16	T	D4
ATTCTGTC	IGVFSM4205CGEZ	080_PWKJ_10F_05	16	R	D5
GCTCTACT	IGVFSM4205CGEZ	080_PWKJ_10F_05	16	T	D5
TGTGTATG	IGVFSM9316HSZQ	081_PWKJ_10M_05	16	R	D6
GTTACGTA	IGVFSM9316HSZQ	081_PWKJ_10M_05	16	T	D6
TCCATTTG	IGVFSM2116DCHR	044_129S1J_10F_05	16	R	D7
CCTGTTGC	IGVFSM2116DCHR	044_129S1J_10F_05	16	T	D7
TTAGCTTC	IGVFSM6420ZYPO	085_PWKJ_10M_05	16	R	D8
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GCTATCAT	IGVFSM4955LORB,IGVFSM9545EZNW	030_AJ_10F_01	16	T	D9
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GCTATCAT	IGVFSM0567BKYP,IGVFSM1552JBSI	080_PWKJ_10F_01	16	T	D9
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GTTTGTGA	IGVFSM6086KVUS,IGVFSM8490BJOU	081_PWKJ_10M_01	16	R	D10
ACATTCAT	IGVFSM6086KVUS,IGVFSM8490BJOU	081_PWKJ_10M_01	16	T	D10
GAAATTAG	IGVFSM3957TLDF,IGVFSM6828FNBA	032_AJ_10F_01	16	R	D11
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GAAATTAG	IGVFSM4249JBEE,IGVFSM7955ABKB	084_PWKJ_10F_01	16	R	D11
TTCGCTAC	IGVFSM4249JBEE,IGVFSM7955ABKB	084_PWKJ_10F_01	16	T	D11
GCAAATTC	IGVFSM1951AMWE,IGVFSM3012KPRJ	035_AJ_10M_01	16	R	D12
CATTCTAC	IGVFSM1951AMWE,IGVFSM3012KPRJ	035_AJ_10M_01	16	T	D12
GCAAATTC	IGVFSM3050OPDO,IGVFSM4510XDYL	085_PWKJ_10M_01	16	R	D12
CATTCTAC	IGVFSM3050OPDO,IGVFSM4510XDYL	085_PWKJ_10M_01	16	T	D12
GAGGTTGA	IGVFSM3021PVNM	036_129S1J_10F_05	16	R	E1
CACTTATC	IGVFSM3021PVNM	036_129S1J_10F_05	16	T	E1
CCTGTCTG	IGVFSM3344MCNS	037_129S1J_10M_05	16	R	E2
ATAAGCTC	IGVFSM3344MCNS	037_129S1J_10M_05	16	T	E2

GTGGGTTC	IGVFSM9049MUBB	038_129S1J_10F_05	16	R	E3
TCATCCTG	IGVFSM9049MUBB	038_129S1J_10F_05	16	T	E3
TTTGCATC	IGVFSM6336XNLQ	041_129S1J_10M_05	16	R	E4
CCTGGTAT	IGVFSM6336XNLQ	041_129S1J_10M_05	16	T	E4
AGGTAATA	IGVFSM7035ACUG	040_129S1J_10F_05	16	R	E5
TGGTATAC	IGVFSM7035ACUG	040_129S1J_10F_05	16	T	E5
GTGCCTTC	IGVFSM5467HYTI	043_129S1J_10M_05	16	R	E6
TTGGGAGA	IGVFSM5467HYTI	043_129S1J_10M_05	16	T	E6
ATGTTTCC	IGVFSM3421MHKE	084_PWKJ_10F_05	16	R	E7
ACTTCATC	IGVFSM3421MHKE	084_PWKJ_10F_05	16	T	E7
CTTAATTC	IGVFSM9876UKZX	045_129S1J_10M_05	16	R	E8
TCTCTAGC	IGVFSM9876UKZX	045_129S1J_10M_05	16	T	E8
TCTGGCTC	IGVFSM4167SEXH,IGVFSM9398YHXY	036_129S1J_10F_01	16	R	E9
ATGCCCTT	IGVFSM4167SEXH,IGVFSM9398YHXY	036_129S1J_10F_01	16	T	E9
TCTGGCTC	IGVFSM7092FDTP,IGVFSM9286SBAR	086_CASTJ_10F_01	16	R	E9
ATGCCCTT	IGVFSM7092FDTP,IGVFSM9286SBAR	086_CASTJ_10F_01	16	T	E9
CATCATTT	IGVFSM3753OVHM,IGVFSM9806TYXF	037_129S1J_10M_01	16	R	E10
CCCAATTT	IGVFSM3753OVHM,IGVFSM9806TYXF	037_129S1J_10M_01	16	T	E10
CATCATTT	IGVFSM4956HIAH,IGVFSM5457PTKY	087_CASTJ_10M_01	16	R	E10
CCCAATTT	IGVFSM4956HIAH,IGVFSM5457PTKY	087_CASTJ_10M_01	16	T	E10
GTTGTCTC	IGVFSM5458IIQG,IGVFSM9530NJWB	038_129S1J_10F_01	16	R	E11
ACTATATA	IGVFSM5458IIQG,IGVFSM9530NJWB	038_129S1J_10F_01	16	T	E11
GTTGTCTC	IGVFSM0690VJRZ,IGVFSM9111THIZ	088_CASTJ_10F_01	16	R	E11
ACTATATA	IGVFSM0690VJRZ,IGVFSM9111THIZ	088_CASTJ_10F_01	16	T	E11
ATCTTCTG	IGVFSM0340DRJV,IGVFSM4269MUAL	041_129S1J_10M_01	16	R	E12
CTCTATAC	IGVFSM0340DRJV,IGVFSM4269MUAL	041_129S1J_10M_01	16	T	E12
ATCTTCTG	IGVFSM6740OVAR,IGVFSM8458TCXX	089_CASTJ_10M_01	16	R	E12
CTCTATAC	IGVFSM6740OVAR,IGVFSM8458TCXX	089_CASTJ_10M_01	16	T	E12
TGTTTGCC	IGVFSM7228AIWD	086_CASTJ_10F_05	16	R	F1
CTGTCTCA	IGVFSM7228AIWD	086_CASTJ_10F_05	16	T	F1
TTCTGTCA	IGVFSM9408IVIR	087_CASTJ_10M_05	16	R	F2
GACCTTTC	IGVFSM9408IVIR	087_CASTJ_10M_05	16	T	F2
ACGGACTC	IGVFSM3996IQCS	088_CASTJ_10F_05	16	R	F3
GATTTGGC	IGVFSM3996IQCS	088_CASTJ_10F_05	16	T	F3
TTTGGTCA	IGVFSM0151EMWS	089_CASTJ_10M_05	16	R	F4

CGTCTAGG	IGVFSM0151EMWS	089_CASTJ_10M_05	16	T	F4
TATCCGGG	IGVFSM6880XYMR	090_CASTJ_10F_05	16	R	F5
TACTCGAA	IGVFSM6880XYMR	090_CASTJ_10F_05	16	T	F5
TGTCATTC	IGVFSM9848ACFP	091_CASTJ_10M_05	16	R	F6
CAGCCTTT	IGVFSM9848ACFP	091_CASTJ_10M_05	16	T	F6
ATTCTCTG	IGVFSM1855TZJL	094_CASTJ_10F_05	16	R	F7
CCTCATTA	IGVFSM1855TZJL	094_CASTJ_10F_05	16	T	F7
TGGCTTCC	IGVFSM8245FNTV	093_CASTJ_10M_05	16	R	F8
CTTATACC	IGVFSM8245FNTV	093_CASTJ_10M_05	16	T	F8
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TCTATTAC	IGVFSM6249RJQV,IGVFSM7479OZRI	040_129S1J_10F_01	16	T	F9
TTGTTGCC	IGVFSM1264GFRJ,IGVFSM5214BQRA	090_CASTJ_10F_01	16	R	F9
TCTATTAC	IGVFSM1264GFRJ,IGVFSM5214BQRA	090_CASTJ_10F_01	16	T	F9
GTCATCTC	IGVFSM6782GARZ,IGVFSM8930SAOK	043_129S1J_10M_01	16	R	F10
CCTGCATT	IGVFSM6782GARZ,IGVFSM8930SAOK	043_129S1J_10M_01	16	T	F10
GTCATCTC	IGVFSM4321DFFH,IGVFSM8357WYSN	091_CASTJ_10M_01	16	R	F10
CCTGCATT	IGVFSM4321DFFH,IGVFSM8357WYSN	091_CASTJ_10M_01	16	T	F10
TTGCTCAT	IGVFSM0539MTAN,IGVFSM6738OHZO	044_129S1J_10F_01	16	R	F11
CAATCCTT	IGVFSM0539MTAN,IGVFSM6738OHZO	044_129S1J_10F_01	16	T	F11
TTGCTCAT	IGVFSM0021WQYA,IGVFSM1257AOMZ	094_CASTJ_10F_01	16	R	F11
CAATCCTT	IGVFSM0021WQYA,IGVFSM1257AOMZ	094_CASTJ_10F_01	16	T	F11
CTGTCTGC	IGVFSM1996CIIR,IGVFSM6742HLKN	045_129S1J_10M_01	16	R	F12
TTGTCTTA	IGVFSM1996CIIR,IGVFSM6742HLKN	045_129S1J_10M_01	16	T	F12
CTGTCTGC	IGVFSM1436ECIH,IGVFSM5017BJEE	093_CASTJ_10M_01	16	R	F12
TTGTCTTA	IGVFSM1436ECIH,IGVFSM5017BJEE	093_CASTJ_10M_01	16	T	F12
TATATTCC	IGVFSM6714CAWV	058_WSBJ_10F_05	16	R	G1
TCACTTTA	IGVFSM6714CAWV	058_WSBJ_10F_05	16	T	G1
ATATTGGC	IGVFSM6424UEBK	057_WSBJ_10M_05	16	R	G2
TGCTTGGG	IGVFSM6424UEBK	057_WSBJ_10M_05	16	T	G2
GTGTCCTC	IGVFSM0803WGCL	060_WSBJ_10F_05	16	R	G3
CGCTCATT	IGVFSM0803WGCL	060_WSBJ_10F_05	16	T	G3
ATCTTCAT	IGVFSM0434YXCE	059_WSBJ_10M_05	16	R	G4
GCCTCTAT	IGVFSM0434YXCE	059_WSBJ_10M_05	16	T	G4
CGTGGTTG	IGVFSM1705RPMT	062_WSBJ_10F_05	16	R	G5
GAGCACAA	IGVFSM1705RPMT	062_WSBJ_10F_05	16	T	G5

TTGCATCC	IGVFSM4873REQN	061_WSBJ_10M_05	16	R	G6
CTCTTAAC	IGVFSM4873REQN	061_WSBJ_10M_05	16	T	G6
TCTTAATC	IGVFSM3444WPKV	096_WSBJ_10F_05	16	R	G7
TCTAGGCT	IGVFSM3444WPKV	096_WSBJ_10F_05	16	T	G7
TGCATTTC	IGVFSM3622HQCN	063_WSBJ_10M_05	16	R	G8
AATTCTGC	IGVFSM3622HQCN	063_WSBJ_10M_05	16	T	G8
GATGTTTC	IGVFSM0966WWIZ,IGVFSM4130KIEV	058_WSBJ_10F_01	16	R	G9
CATTCTCA	IGVFSM0966WWIZ,IGVFSM4130KIEV	058_WSBJ_10F_01	16	T	G9
GATGTTTC	IGVFSM1078KTYU,IGVFSM8134HPUH	092_CASTJ_10F_01	16	R	G9
CATTCTCA	IGVFSM1078KTYU,IGVFSM8134HPUH	092_CASTJ_10F_01	16	T	G9
ATCTTGTC	IGVFSM4107ZNSE,IGVFSM6703AANL	047_NZOJ_10M_01	16	R	G10
ACTTGCCT	IGVFSM4107ZNSE,IGVFSM6703AANL	047_NZOJ_10M_01	16	T	G10
ATCTTGTC	IGVFSM7450WJIV,IGVFSM8498UCVA	057_WSBJ_10M_01	16	R	G10
ACTTGCCT	IGVFSM7450WJIV,IGVFSM8498UCVA	057_WSBJ_10M_01	16	T	G10
TCATATTC	IGVFSM0298LFOQ,IGVFSM8754ELWG	048_NZOJ_10F_01	16	R	G11
ATCATTGC	IGVFSM0298LFOQ,IGVFSM8754ELWG	048_NZOJ_10F_01	16	T	G11
TCATATTC	IGVFSM0752ZRMG,IGVFSM3371TFMC	060_WSBJ_10F_01	16	R	G11
ATCATTGC	IGVFSM0752ZRMG,IGVFSM3371TFMC	060_WSBJ_10F_01	16	T	G11
TGGCCTCT	IGVFSM2572HXXE,IGVFSM9197PRMM	049_NZOJ_10M_01	16	R	G12
GTTCAACA	IGVFSM2572HXXE,IGVFSM9197PRMM	049_NZOJ_10M_01	16	T	G12
TGGCCTCT	IGVFSM2287EHGA,IGVFSM9356QKBM	059_WSBJ_10M_01	16	R	G12
GTTCAACA	IGVFSM2287EHGA,IGVFSM9356QKBM	059_WSBJ_10M_01	16	T	G12
CGTTGTCT	IGVFSM9090ZNRS	046_NZOJ_10F_05	16	R	H1
CCATTTGC	IGVFSM9090ZNRS	046_NZOJ_10F_05	16	T	H1
TCTTGTCA	IGVFSM4130EMOU	047_NZOJ_10M_05	16	R	H2
GACTTTGC	IGVFSM4130EMOU	047_NZOJ_10M_05	16	T	H2
TATTCCTG	IGVFSM1384SGZS	048_NZOJ_10F_05	16	R	H3
ATTGGCTC	IGVFSM1384SGZS	048_NZOJ_10F_05	16	T	H3
TCCATGTC	IGVFSM6644MYFA	049_NZOJ_10M_05	16	R	H4
GTGCTAGC	IGVFSM6644MYFA	049_NZOJ_10M_05	16	T	H4
TTGTCATC	IGVFSM9429JZJC	050_NZOJ_10F_05	16	R	H5
CTTTCAAC	IGVFSM9429JZJC	050_NZOJ_10F_05	16	T	H5
ATTTCCTG	IGVFSM1239ETDH	051_NZOJ_10M_05	16	R	H6
ACTATTGC	IGVFSM1239ETDH	051_NZOJ_10M_05	16	T	H6
GTGTCTCC	IGVFSM3658ESYZ	052_NZOJ_10F_05	16	R	H7

ACTGGCTT	IGVFSM3658ESYZ	052_NZOJ_10F_05	16	T	H7
GTGTGTGT	IGVFSM7821JHFX	053_NZOJ_10M_05	16	R	H8
ATTAGGCT	IGVFSM7821JHFX	053_NZOJ_10M_05	16	T	H8
TATGCTTC	IGVFSM2120LYVH,IGVFSM8205AVFV	050_NZOJ_10F_01	16	R	H9
GCCTTTCA	IGVFSM2120LYVH,IGVFSM8205AVFV	050_NZOJ_10F_01	16	T	H9
TATGCTTC	IGVFSM2047VYYJ,IGVFSM7560RVGB	062_WSBJ_10F_01	16	R	H9
GCCTTTCA	IGVFSM2047VYYJ,IGVFSM7560RVGB	062_WSBJ_10F_01	16	T	H9
ATGGTGTT	IGVFSM3556PAEN,IGVFSM9711JMMI	051_NZOJ_10M_01	16	R	H10
ATTCTAGG	IGVFSM3556PAEN,IGVFSM9711JMMI	051_NZOJ_10M_01	16	T	H10
ATGGTGTT	IGVFSM3364FXFB,IGVFSM4632PCNC	061_WSBJ_10M_01	16	R	H10
ATTCTAGG	IGVFSM3364FXFB,IGVFSM4632PCNC	061_WSBJ_10M_01	16	T	H10
GAATAATG	IGVFSM5374RXFR,IGVFSM9813BNZU	052_NZOJ_10F_01	16	R	H11
CCTTACAT	IGVFSM5374RXFR,IGVFSM9813BNZU	052_NZOJ_10F_01	16	T	H11
GAATAATG	IGVFSM1348QFIK,IGVFSM6620ZEBK	096_WSBJ_10F_01	16	R	H11
CCTTACAT	IGVFSM1348QFIK,IGVFSM6620ZEBK	096_WSBJ_10F_01	16	T	H11
CCTCTGTG	IGVFSM3167TBKE,IGVFSM9817PYLC	053_NZOJ_10M_01	16	R	H12
ACATTTGG	IGVFSM3167TBKE,IGVFSM9817PYLC	053_NZOJ_10M_01	16	T	H12
CCTCTGTG	IGVFSM7877XJMN,IGVFSM9074ZTED	063_WSBJ_10M_01	16	R	H12
ACATTTGG	IGVFSM7877XJMN,IGVFSM9074ZTED	063_WSBJ_10M_01	16	T	H12

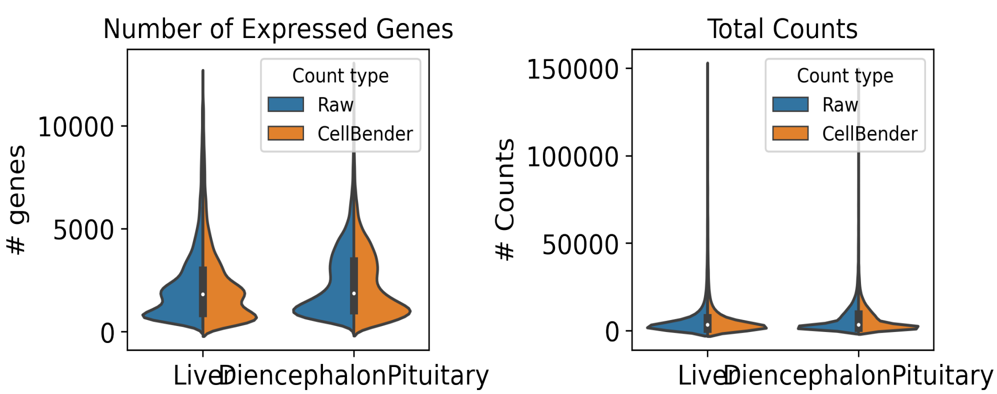
6. Description of data processing workflow and h5ad files

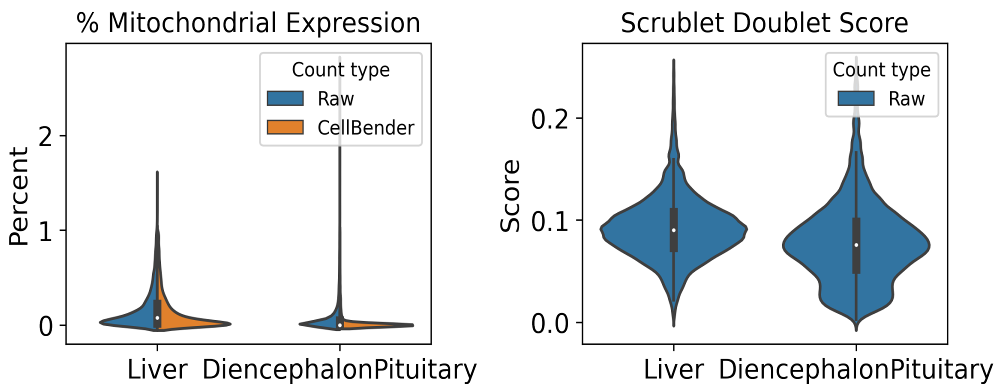
Data from Subpool_3 were combined with all other subpools into one AnnData object per tissue as well as matching data from other plates for processing and cell type annotation. The AnnData contains detailed sample and cell metadata in the observation (obs) table and gene information in the variables table (var). Cell IDs (adata.obs.index) are re-formatted to be human-readable and unique across subpools and experiments by appending subpool and experiment IDs. The final AnnData contains both raw and CellBender denoised counts matrices in separate layers: adata.layers['raw_counts'] and adata.layers['cellbender_counts']. The current adata.X points to the CellBender denoised counts. Cells or nuclei are filtered by >0.5 cell_probability from CellBender analysis.

In addition to CellBender filtering, nuclei were also filtered by the following QC parameters:

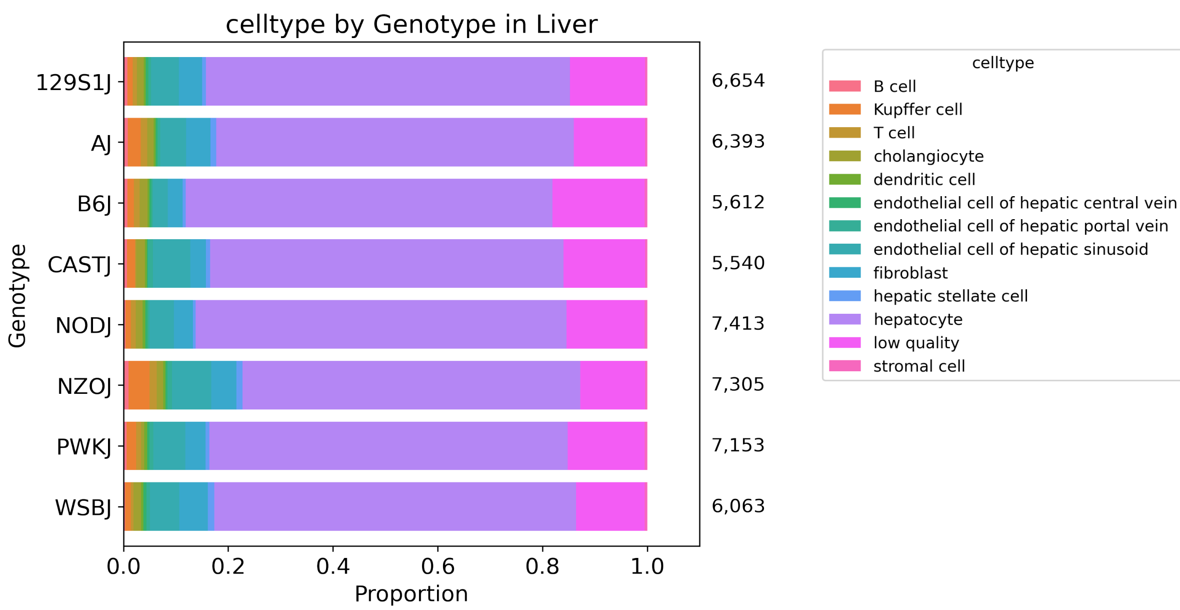
Table 8: QC thresholds

Parameter	Threshold
Minimum # UMIs	500.0
Maximum # UMIs	150000.0
Minimum # genes	250.0
Maximum % mito.	1.0
Maximum doublet score	0.25





After filtering, data for each tissue was normalized and clustered using CellBender denoised counts. Normalization includes regression of technical parameters (number of genes expressed per cell and percent mitochondrial gene expression). Harmony was used to integrate exome capture and non-exome capture subpools for annotation. A Leiden clustering resolution of 1 was used for 53 total clusters in Liver and 55 clusters in DiencephalonPituitary, using all tissue data across experiments. Cell type annotations were assigned per Leiden cluster or subcluster (leiden_R) using expression of canonical cell type marker genes.



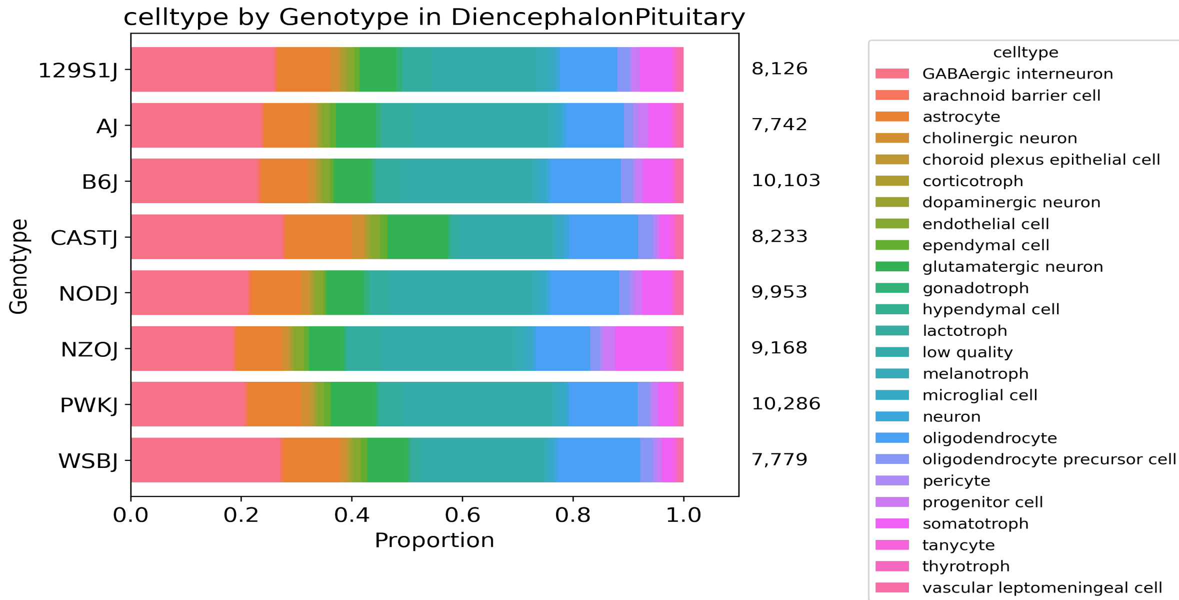


Table 9: Description of obs keys in h5ad file

Key	Description
cellID	Index of obs table. Includes barcoding wells for each round (1_2_3), subpool, and experiment.
lab_sample_id	Human-readable sample name used in-house.
sample	IGVF tissue accession.
plate	Experiment ID, e.g. igvf_003.
subpool	Subpool ID, e.g. Subpool_2. Number corresponds to Illumina index ID from Parse Bio kit.
SampleType	Nuclei or Cells.
Tissue	Human-readable tissue name, e.g. Kidney.
Sex	Male or Female.
Age	Age in postnatal months (PNM) of the mouse, e.g. PNM_02.
Genotype	Mouse genotype/strain, e.g. B6J.
leiden	Numeric Leiden cluster starting from 0.
leiden_R	If necessary, Leiden clusters and any sub-clusters, e.g. 15_1.
subpool_type	Exome capture (EX) or no exome capture (NO).
general_celltype	Broadest level of cell type annotations, e.g. neuron.
general_CL_ID	EMBL CL database ID for general cell type annotation, e.g. CL:0000540.
celltype	Finest level of cell type annotation with a CL ID, e.g. Vip+ GABAergic cortical interneuron.
CL_ID	EMBL CL database ID for cell type annotation, CL:4023016
subtype	Finest level of cell type annotation that may not have a CL ID. Mostly matches celltype.
Protocol	Cell barcoding and library building protocol/kit, e.g. Parse_WT.

Chemistry	Protocol chemistry version, v2 or v3.
bc	24-nucleotide sequence corresponding to 3 8-nucleotide barcode rounds (3, 2, 1).
bc1_sequence	8-nucleotide sequence barcode for round 1.
bc2_sequence	8-nucleotide sequence barcode for round 2.
bc3_sequence	8-nucleotide sequence barcode for round 3.
bc1_well	Round 1 well.
bc2_well	Round 2 well.
bc3_well	Round 2 well.
Row	Sample barcoding plate (round 1) row.
Column	Sample barcoding plate (round 1) column.
well_type	Single or Multiplexed.
Mouse_Tissue_ID	Same as lab_sample_id.
Multiplexed_sample1	Sample 1 from multiplexed well (if applicable)
Multiplexed_sample2	Sample 2 from multiplexed well (if applicable)
DOB	Date of birth of the mouse in month/day/year format.
Age_days	Age of the mouse in days.
Body_weight_g	Body weight in grams of the mouse.
Estrus_cycle	Estimated estrus stage of the mouse, if applicable.
Dissection_date	Date the tissue was harvested in month/day/year format.
Dissection_time	Time in hours:minutes (24-hour time) the mouse was sacrificed.
ZT	Number of hours into the light cycle, from lights on (06:30, ZT0) to lights off (20:30, ZT14).
Dissector	Initials of technician who dissected the tissue.
Tissue_weight_mg	Tissue weight in milligrams.
Notes	Notes taken during sample collection, experiment, or data processing.
n_genes_by_counts_raw	The number of genes with at least 1 count, calculated using raw counts.
total_counts_raw	Total counts (UMIs), calculated using raw counts.
total_counts_mt_raw	Total counts for mitochondrial genes, calculated using raw counts.
pct_counts_mt_raw	Percent of total counts in mitochondrial genes, calculated using raw counts.
n_genes_by_counts_cb	The number of genes with at least 1 count, calculated using CellBender counts.
total_counts_cb	Total counts (UMIs), calculated using CellBender counts.
total_counts_mt_cb	Total counts for mitochondrial genes, calculated using CellBender counts.
pct_counts_mt_cb	Percent of total counts in mitochondrial genes, calculated using CellBender counts.
doublet_score	Scrublet doublet score, calculated using CellBender counts.
predicted_doublet	Scrublet predicted doublet (True or False).
background_fraction	Fraction of counts CellBender predicted to be background noise.

cell_probability	CellBender probability that the cell is real.
cell_size	CellBender size factor.
droplet_efficiency	CellBender statistic indicating transcript capture efficiency per cell.
B6J_klue_counts	Estimated counts for B6J genotype from klue analysis.
NODJ_klue_counts	Estimated counts for NODJ genotype from klue analysis.
AJ_klue_counts	Estimated counts for AJ genotype from klue analysis.
PWKJ_klue_counts	Estimated counts for PWKJ genotype from klue analysis.
129S1J_klue_counts	Estimated counts for 129S1J genotype from klue analysis.
CASTJ_klue_counts	Estimated counts for CASTJ genotype from klue analysis.
WSBJ_klue_counts	Estimated counts for WSBJ genotype from klue analysis.
NZOJ_klue_counts	Estimated counts for NZOJ genotype from klue analysis.

7. Package versions

Table 10: Package versions

package	version
kallisto	0.50.1
bustools	0.43.2
kb	0.28.2
Python	3.9.0
snakemake	7.32.0
cellbender	0.3.2
scanpy	1.10.2
scrublet	0.2.3
anndata	0.10.8
pandas	2.2.2
numpy	1.26.4
harmonypy	0.0.9

8. Contact information and GitHub

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Pipeline repo: https://github.com/mortazavilab/parse_pipeline

Code used to make this report: https://github.com/mortazavilab/8cube_manuscript/blob/main/plate_report/Subpool_report_igvf_005.ipynb

Code used to process and annotate the data for main tissue:
https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/Liver.ipynb

Code used to process and annotate the data for multiplexed tissue: https://github.com/mortazavilab/8cube_manuscript/blob/main/annotation/DiencephalonPituitary.ipynb